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To the Editors-in-Chief
Journal of Intelligent Information Systems

Dear Editors,

I submit the manuscript **"AURA-HOI: High-Occupancy Itemset Mining via Equivalence Classes and Residual Envelope Pruning"** for consideration as a research article in the *Journal of Intelligent Information Systems*.

Problem. Occupancy measures how much of its supporting transactions an itemset covers. It is not anti-monotone, so exact miners of high-occupancy itemsets prune the search with upper bounds. The manuscript shows that the bound used by the existing exact miners depends only on transaction lengths and reduces to the support when all transactions have the same length, a common situation in categorical data; these miners then enumerate every frequent itemset.

Contributions.

- A length-capped residual bound that combines transaction lengths with the largest itemset that can still be generated in a branch, with proofs that it is safe and never weaker than the two bounds it combines.
- AURA-HOI, a depth-first miner over equivalence classes of transaction-ID lists, with a proof that it returns exactly the set of high-occupancy itemsets, and a minimal counterexample showing that a natural shortcut in the residual computation loses results.
- A reproducible evaluation: verification against an exact rational-arithmetic oracle, output parity with two exact baselines (HEP and DFHOI) on nine public benchmarks, repeated single-threaded runs without a time limit, memory measured as the increase attributable to mining, an ablation study, a scalability study and a signed-rank test. The results also report where the method does not help.

Fit with the journal. JIIS publishes work at the intersection of artificial intelligence and data management, including data mining and knowledge discovery. The manuscript contributes an exact, provably correct pattern-mining algorithm and a verification protocol for comparing exact miners.

Prior submission. This manuscript is a revised version of a manuscript submitted to JIIS earlier this year (Manuscript ID 8f0df156-2539-436c-8fc5-f7878f5b3403), which I withdrew before peer review after finding errors in it. The method has since been corrected and proved, and all experiments have been repeated under the protocol described in the manuscript.

Declarations. The manuscript is original, has not been published, and is not under consideration elsewhere. The research received no funding, and the author declares no competing interests.

Sincerely,

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